

CECS 381: Stochastic Computing

Programming Assignment 2 (Units 6–9: Monte Carlo, Markov Chains & PageRank)

Summer 2026 Assigned: August 4 Due: August 13, 2026 (11:59 PM)

California State University, Long Beach

Name: _____ Student ID: _____

Instructions. Submit a single Jupyter notebook (.ipynb) plus a PDF export. Use the Python standard library, `numpy`, and `matplotlib`. Your code must run top to bottom without errors. Answer the written questions in markdown cells. Total: 100 points.

Question:	1	2	3	Total
Points:	30	35	35	100
Score:				

1. (30 points) **Monte Carlo estimation.** Use random sampling to estimate a constant and an integral, and study the error rate.
 - (a) (10 pts) Estimate π by the dart method: sample points in the unit square and count the fraction with $x^2 + y^2 \leq 1$. Report estimates for $n = 10^2, 10^4, 10^6$.
 - (b) (10 pts) Estimate $\int_0^1 e^{-x^2} dx$ by Monte Carlo and compare with the true value ≈ 0.7468 .
 - (c) (10 pts) Plot $|\hat{\pi} - \pi|$ versus n on log–log axes and overlay a reference line proportional to $1/\sqrt{n}$. Comment (markdown) on the observed convergence rate.
2. (35 points) **Markov chain simulation and the stationary distribution.** Consider the server chain with states Up (0) and Down (1) and transition matrix $P = \begin{pmatrix} 0.8 & 0.2 \\ 0.6 & 0.4 \end{pmatrix}$.
 - (a) (15 pts) Simulate the chain for $T = 10^6$ steps starting from Up, recording the fraction of time spent in each state.
 - (b) (12 pts) Compute the stationary distribution by power iteration ($\pi \leftarrow \pi P$ repeated) and confirm it matches the empirical fractions.
 - (c) (8 pts) Compute the stationary distribution a second way, as the left eigenvector of P with eigenvalue 1, and verify all three agree.
3. (35 points) **PageRank by power iteration.** Implement PageRank on the 4-page web from Class 20: A links to B and D; B links to C; C links to A and B; D links to C.
 - (a) (12 pts) Build the raw link matrix H (rows sum to 1), then form the Google matrix $G = \alpha H + (1 - \alpha) \frac{1}{N} \mathbf{1}$ with $\alpha = 0.85$.
 - (b) (15 pts) Run power iteration from the uniform vector until convergence, and print the PageRank of each page.
 - (c) (8 pts) Report the ranking. Which page wins, and why (markdown)? Does link *quality* or *quantity* explain the result?